

Arduino Capstone: Design Your Own IoT Gadget

Learn How Sensors, Brains, and Actuators Work Together to Solve Real-World Problems!

1. Real-World Problem: Water Tank Overflow

Imagine leaving a water pump running and having an overhead tank overflow! How can we build an intelligent IoT gadget to solve this exact problem?

Solution Concept: Smart Water Level Alarm

An **Ultrasonic Sensor** checks the water level in real-time. The **Arduino UNO** processes the data and determines if the tank is full. If water rises too high, a **Buzzer** rings immediately to alert you to turn off the pump!

2. What Makes an IoT Gadget? (Anatomy of a Smart Device)

Every IoT (Internet of Things) gadget is built using three essential pillars that mimic human body functions:

Eyes = Sensors

Sensors detect physical conditions around them in the real world.

- **Ultrasonic Sensor:** Measures distance using sound.
- **LDR (Light Dependent Resistor):** Detects light/darkness.
- **PIR Sensor:** Detects motion and movement.
- **Mitti Moisture Sensor:** Measures soil dampness.

Brain = Arduino UNO

The microcontroller board receives data, processes logic, and decides what to do based on coded rules.

Hands = Actuators

Actuators take physical actions in the environment!

- **Buzzer:** Produces sound alerts.
- **LEDs:** Provide visual light signals.
- **LCD Screen:** Displays text messages and status.

3. How an IoT Gadget Works (3-Step Loop)

Step 1: A Sensor Spots the Problem — Reads environmental data (e.g., distance, light level, motion).

Step 2: Arduino UNO Decides What to Do — Compares sensor input against predefined thresholds in code.

Step 3: An Actuator Takes Action! — Triggers a response like sounding an alarm or turning on a light.

4. Capstone Blueprint Tracks

Choose your capstone project track and design your 1-page blueprint before building:

Track A: Smart Jal-Rakshak (Water Overflow Alarm)

Uses an ultrasonic distance sensor to detect rising water levels and triggers a buzzer alarm when water is near the top of the tank.

Track B: Auto-Plant Watering System

Uses a Mitti (soil) moisture sensor to detect dry soil and triggers a pump or LED indicator to signal when plants need water.

Track C: Smart Colony Security Light

Uses a PIR motion sensor to detect movement in dark corridors and automatically switches on a high-brightness security light.



Builder Mindset: Mistakes = Adventures!

When prototyping in **Tinkercad** or with real hardware, trial and error is normal. Troubleshooting wiring and adjusting thresholds in code is how true engineers learn!

Student Practice Sheet

Arduino Capstone: IoT Gadget Principles & Components

Name: _____

Date: _____

Q1. What are the three main components that make up an IoT gadget, and what human body parts are they compared to?

Q2. Which of the following devices is classified as a sensor? (Circle the correct answer)

A) Buzzer B) Ultrasonic Sensor C) LED

Q3. Explain the role of an actuator in an IoT system and give two examples of actuators.

Q4. If an LDR (Light Dependent Resistor) sensor detects darkness, what action could the Arduino UNO instruct an actuator to take?

Q5. Describe the step-by-step working loop of the "Smart Jal-Rakshak" water overflow alarm system when a tank fills up.

Q6. Which project track uses a Mitti (Soil) Moisture Sensor, and what real-life problem does it solve?

Q7. Capstone Challenge: Imagine you want to design a "Smart Pet Feeder" IoT gadget. Identify which sensor you would use for input, what decision the Arduino should make, and which actuator would perform the action.
